

Vector Calculus Prerequisites

Please review:

1. **Vector algebra:** vector addition, dot product, cross product.
2. **Del operator:** gradient, divergence, curl of vector fields.
3. **Curvilinear coordinates:** cylindrical and spherical coordinates.
4. **Integral calculus:** line/surface/volume integrals; Divergence theorem, Stokes' theorem, Green's theorem.

1 Coulomb's Law and Electric Force

Coulomb's law (experimental law) gives the electrostatic force between two point charges Q_1 and Q_2 . Let their position vectors be $\vec{r}_1$ and $\vec{r}_2$. Define

$$\vec{R}_{12} = \vec{r}_2 - \vec{r}_1, \quad R_{12} = \|\vec{R}_{12}\|, \quad \hat{\mathbf{a}}_{12} = \frac{\vec{R}_{12}}{R_{12}}.$$

Then the force on Q_2 due to Q_1 is

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{R_{12}^2} \hat{\mathbf{a}}_{12}$$

and $\vec{F}_{21} = -\vec{F}_{12}$.

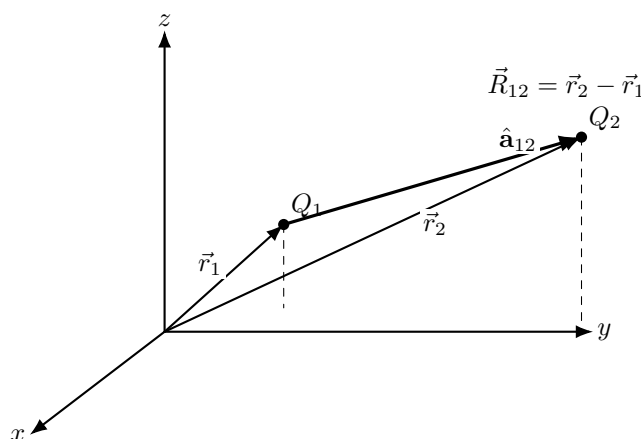


Figure 1: Geometry for Coulomb's law: $\vec{R}_{12} = \vec{r}_2 - \vec{r}_1$, $\hat{\mathbf{a}}_{12} = \vec{R}_{12}/\|\vec{R}_{12}\|$.

SI Constant

$$k = \frac{1}{4\pi\epsilon_0}, \quad \epsilon_0 \approx 8.854 \times 10^{-12} \text{ F m}^{-1}.$$

Key Takeaways

- Coulomb force acts along the line joining the charges: direction is $\hat{\mathbf{a}}_{12}$.
- Magnitude scales as $1/R^2$ and depends on the product Q_1Q_2 .
- Newton's 3rd law holds: $\vec{F}_{21} = -\vec{F}_{12}$.

2 How Powerful is the Coulomb Force? (Feynman Thought Experiment)

Assume a 100 kg human body (density $\approx 1 \text{ g/cm}^3$) has about 1% excess electrons. A rough electron number density is $\sim 10^{23} \text{ electrons/cm}^3$.

Back-of-the-envelope estimate

1% of 100 kg $\Rightarrow$ 1 kg excess electrons.

Volume of 1 kg (1000 g) at 1 g/cm^3 is 10^3 cm^3 , hence

$$N \approx 10^3 \times 10^{23} = 10^{26} \text{ electrons.}$$

Charge magnitude:

$$q \approx 10^{26} e \approx 10^{26} \times 1.6 \times 10^{-19} \text{ C} \approx 1.6 \times 10^7 \text{ C.}$$

If two such bodies are 1 m apart,

$$F \approx k \frac{q^2}{r^2} \approx (9 \times 10^9) \frac{(1.6 \times 10^7)^2}{1^2} \text{ N} \approx 2.3 \times 10^{24} \text{ N,}$$

which corresponds to an equivalent “weight” of $\sim 10^{23} \text{ kg}$ (using $g \approx 10 \text{ m/s}^2$). This electrostatic force of repulsion can lift the earth!

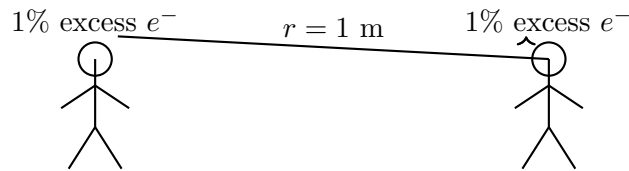


Figure 2: Feynman-style illustration: tiny charge imbalance can lead to enormous Coulomb forces.

Key Takeaways

- Even a **tiny charge imbalance** can create **enormous** forces.
- Macroscopic objects remain **nearly neutral** because Coulomb forces are so strong.
- Back-of-the-envelope estimates are useful to build physical intuition.

3 Electric vs Gravitational Force Between Two Electrons

Two electrons separated by distance d :

$$|F_g| = G \frac{m_e^2}{d^2}, \quad |F_e| = k \frac{e^2}{d^2}.$$

Ratio (independent of d):

$$\boxed{\frac{|F_e|}{|F_g|} = \frac{ke^2}{Gm_e^2} \sim 10^{43}}$$

(using $e \approx 1.6 \times 10^{-19}$ C, $\varepsilon_0 \approx 8.85 \times 10^{-12}$, $G \approx 6.7 \times 10^{-11}$ N m²/kg², $m_e \approx 9.1 \times 10^{-31}$ kg).

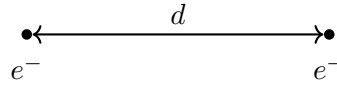


Figure 3: Two electrons separated by distance d . Gravity attracts, but electrostatic force dominates and repels.

Heavier particle: protons

For protons, $m_p \approx 2000 m_e$ and $|q_p| = e$:

$$\left. \frac{|F_e|}{|F_g|} \right|_{\text{protons}} \approx \frac{10^{43}}{(2000)^2} \approx 10^{37}.$$

(As quoted in class: electricity is “billion billion billion billion” times stronger than gravity.)

Key Takeaways

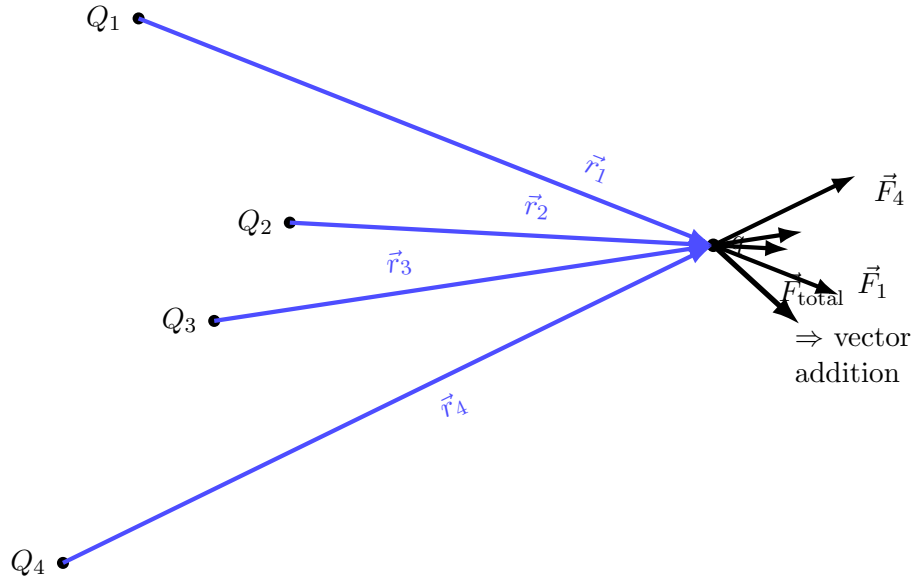
- The ratio $\frac{F_e}{F_g}$ is **independent of distance** (both scale as $1/d^2$).
- For electrons, EM dominates gravity by about 10^{42} – 10^{43} .
- Gravity becomes relevant mainly at **astronomical** mass scales, not atomic scales.



4 Superposition Principle

For a set of source charges $\{Q_i\}$, the net force on a test charge q is the vector sum:

$$\vec{F}_{\text{total}} = \sum_{i=1}^N \vec{F}_i = \frac{q}{4\pi\epsilon_0} \sum_{i=1}^N \frac{Q_i}{r_i^2} \hat{\mathbf{a}}_i$$



Precautionary note: superposition may fail in some media

Superposition assumes the presence of other charges does not alter the “interaction law” between a given source and the test charge. If the **medium properties change** due to other charges (e.g., $\epsilon = \epsilon(\text{field})$ or $\epsilon = \epsilon(Q_2)$), then the force between Q_1 and q can change in the presence of Q_2 .

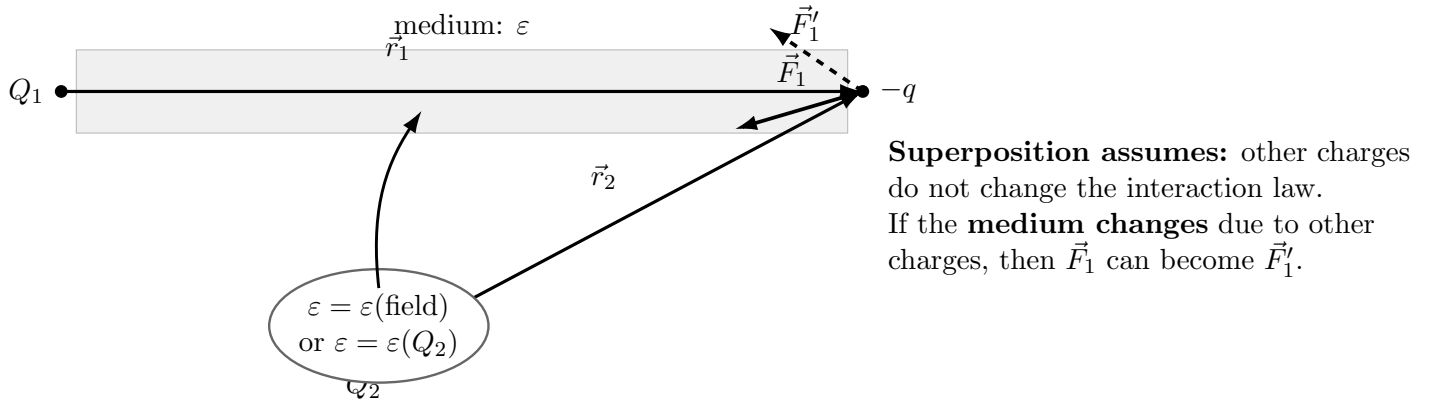


Figure 4: Failure of superposition in nonlinear or charge-dependent media: adding Q_2 can modify the medium permittivity and hence alter the force between Q_1 and q .

Key Takeaways

- Forces and fields add by **vector superposition**: $\vec{F} = \sum \vec{F}_i$, $\vec{E} = \sum \vec{E}_i$.
- Superposition is valid in **linear media** and vacuum.
- It can fail when the medium response becomes **nonlinear** / **field-dependent** (e.g., $\epsilon = \epsilon(\vec{E})$).

5 Superposition Example: Force Due to a Dipole (Radial Component Outline)

Consider a dipole with charges $+Q$ and $-Q$ separated by distance d . A test charge q is located at point P . Let the distances from $+Q$ and $-Q$ to P be r_1 and r_2 .

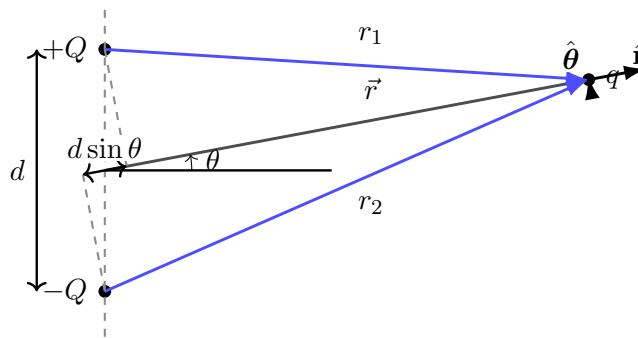


Figure 5: Dipole geometry ($d \ll r_1, r_2$). With θ measured from the $+x$ axis and dipole separation along y , the projected separation along the observation direction is $d \sin \theta$, giving $r_2 \approx r_1 + d \sin \theta$.

Along the radial direction,

$$F_r = \frac{Qq}{4\pi\epsilon_0} \left(\frac{1}{r_1^2} - \frac{1}{r_2^2} \right).$$

Using the geometric approximation used in class,

$$r_2 \approx r_1 + d \sin \theta, \quad d \ll r_1,$$

$$\frac{1}{(r_1 + d \sin \theta)^2} \approx \frac{1}{r_1^2} \frac{1}{1 + 2(d \sin \theta)/r_1} \approx \frac{1}{r_1^2} \left(1 - \frac{2d \sin \theta}{r_1} \right),$$

which yields (to first order)

$$F_r \approx \left(\frac{Qq}{4\pi\epsilon_0 r_1^2} \right) \left(\frac{2d \sin \theta}{r_1} \right).$$

Exercise: similarly derive F_θ .

Key Takeaways

- A dipole is two equal and opposite charges separated by small d .
- For $d \ll r$, use **first-order approximations** to simplify expressions.
- Dipole fields decay faster than monopole fields: typically $\sim 1/r^3$ in electrostatics.

The Force Paradox:

Why is the Universe Ruled by its Weakest Force?

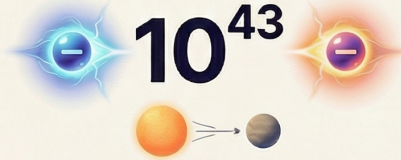
THE PROBLEM: AN UNIMAGINABLY STRONG FORCE

Coulomb's Law:
The Force Between Charges



The force is proportional to the charges and inversely proportional to the distance squared.

10^{43} Times Stronger Than Gravity



The electric repulsion between two electrons is 43 orders of magnitude stronger than their gravitational attraction.

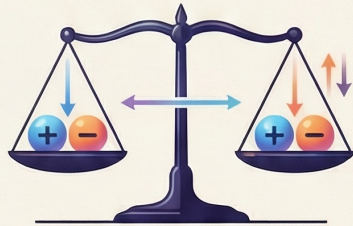
A 1% charge imbalance in two people could create a force strong enough to lift the Earth.

This illustrates the immense power hidden within ordinary, electrically neutral matter.



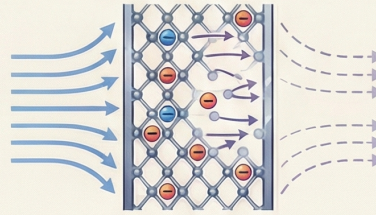
THE SOLUTION: THE GREAT CANCELLATION

The Cancellation Effect



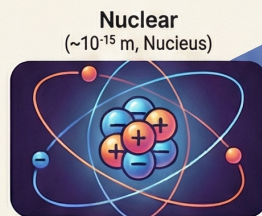
Unlike gravity which only attracts, the existence of positive and negative charges causes electric forces to cancel out.

Shielding Causes Exponential Decay

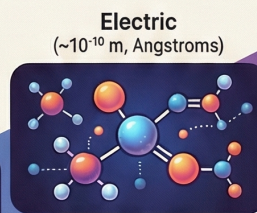


Mobile charged particles in a medium (like gas or a conductor) rearrange to neutralize electric fields.

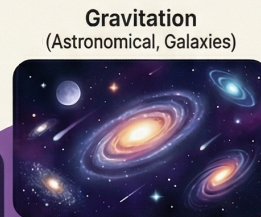
Strongest Forces Dominate Smallest Scales



Governs The Formation Of Elements



Governs The Formation Of Chemistry & Molecules



Governs The Formation Of Universe Structure

The strongest forces cancel out first, allowing weaker, cumulative forces to dominate over vast distances.

NotebookLM



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